

DEPARTMENT	ELECTRONICS AND COMMUNICATION						
Course Code	24EC111	Total Credits	3	Course Type	Engineering Science Course		
Course Title	Elements of Electronics Engineering						
Teaching Learning Process		Contact Hours	Credits	Assessment in Weightage and marks			
	Lecture	39	3		CIE	SEE	Total
	Tutorial	0	0	Weightage	40 %	60 %	100 %
	Practical	0	0	Maximum Marks	40 Marks	60 Marks	100 Marks
	Total	39	3	Minimum Marks	20 marks	25 marks	45 Marks

Note: For passing the student has to score a minimum of 45 marks (CIE+SEE: 20+25 or 21+24)

COURSE PREREQUISITE: Basic Principles Physics.

COURSE OBJECTIVE: To understand the Fundamental Concepts of Principles of Electronics Engineering.

COURSE OUTCOMES (COs): After completing this course, students should be able to

CO#	Course Outcomes	Highest Level of Cognitive Domain
CO1	Analyze the working of various analog electronics circuits and explore its applications.	L2
CO2	Apply the digital logic to build the combinational and sequential circuits.	L3
CO3	Design the circuit using ICs for the given specifications (741, 723, 555).	L2
CO4	Explain the principles of communication	L2
CO5	Demonstrate the skill sets using software tools for simulation of electronics circuit.	L3

L1 – Remember, L2 – Understand, L3 – Apply, L4 – Analyze, L5 – Evaluate, L6 - Create

UNIT No.	Content	Hours	
		Lecture	Tutorial
1	Unit-1: Semiconductor Diodes Diode: Construction and working, Characteristics (ideal and practical), Diode resistances, Load line analysis. Applications of Diode: Rectifiers – Half wave, Full wave (Center tapped and Bridge) with and without C filter (only qualitative analysis), Basic Clipper and Clamper circuits. Zener diode and Photo diode: Construction, Working and Characteristics.	8	0
2	Unit-2: Transistors BJT: Construction and working, Common Emitter Characteristics with load line analysis. Applications: Switch and Amplifier JFET: Construction, Working and Characteristics. MOSFET: N type and P type (Depletion and Enhancement Mode) Construction, Working and Characteristics. Applications: NOT, NAND, NOR gates using CMOS	8	0
3	Unit-3: Digital Electronics Basic and universal Logic Gates, simplification and realization of Boolean expression using basic gates. Combinational logic: Half adder, Full adder, Half subtractor, Full subtractor, 2:1 and 4:1 MUX, 1:2 and 1:4 DEMUX, 1Bit Magnitude comparator. Sequential logic: Latch, Flip flops (SR, JK, D, T), 4 bit Synchronous and Asynchronous up counter, 4bit Shift registers.	8	0
4	Unit 4: Integrated circuits Differential amplifier: Operation with respect to common mode and differential inputs. Op-amps: Block diagram, Ideal and Practical characteristics, Inverting and Non inverting amplifier, Saturable property. Applications of Op-amp: Voltage follower, Adder, Subtractor, Integrator and Differentiator, 78XX IC regulators, Digital IC (TTL Nand Gate), 555 Timer Circuit (Astable Multivibrator).	8	0
5	Unit-5: Communication Basics EM spectrum and Applications, Block diagram of Analog and Digital Communication System, Need for Modulation, AM and FM, ASK and FSK (Qualitative analysis only), Principles of Mobile communication- Wireless communication systems and Cellular hierarchy. Basics of Computer Communication-OSI layer, Network topology, MODEM.	7	0

Text Books:

1. Robert Boylestad: "Electronic Devices and circuit theory", 11 Edition, Pearson, 2015.
2. Ramakant A. Gayakwad, "Op-Amps and Linear Integrated Circuits", 4th Edition, Pearson, 2015

Reference Books:

1. Floyd and Jain, "Digital Fundamentals", 11 Edition, Pearson, 2017.
2. B.P. Lathi and Zhi Ding, "Modern Digital And Analog Communication Systems: 4th Edition (Paperback) South Asia edition, Oxford 2017.

Web/Digital resources:

- 1 EBook: <http://www.freebookcentre.net/Electronics/Communication-Books.html>
- 2 EBook: <http://www.freebookcentre.net/Electronics/Devicesandequipments-Books.html/>
- 3 Video-Lecture: <http://nptel.ac.in/courses/108102095/>

SWAYAM/NPTEL:

- 1 EBook: <http://www.freebookcentre.net/Electronics/Communication-Books.html>
- 2 EBook: <http://www.freebookcentre.net/Electronics/Devicesandequipments-Books.html>
- 3 Video Lecture: <http://nptel.ac.in/courses/108102095/>

PRACTICE BASED LEARNING:

No	Topics to be covered	Tools and Techniques	Expected Skill/Ability
1	Circuit design and implementation (mini project)	Any electronics simulation software (LAB view, TINA TI, Multisim, etc.,	Familiar with electronic component and devices and its working

OR**Laboratory Experiments: (For Circuit Branch Students)**

1. Diode Characteristics
2. Rectifiers- Half wave/ Full Wave
3. Common Emitter Configuration Characteristics
4. Operational Amplifier Circuits – 1
5. Operational Amplifier Circuits – 2
6. Basic Digital Electronics- Logic gates, SOP, POS
7. Adders and Subtractors
8. Mux/ Demux Using 74153 & 74139
9. Flip Flops- SR, JK, D, T
10. Shift Register

Self-Learning Exercises:

1. Case Study projects
2. Mini Projects
3. Any others

Course Articulation:

COURSE OUTCOMES ↓	PROGRAM OUTCOMES											
	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	3									
CO2	3	3	3									
CO3	3	3	3									
CO4	3	3	3									
CO5	3	3	3	2	3				2	2		

High – 3, Medium – 2, Low – 1